SMART WASTE SEGRIGATION MONITORING

ABSTRACT

The Smart Waste Segregation Monitoring System is an IoT-based solution developed using Arduino Uno to improve waste management efficiency. The system uses a Moisture Sensor to identify wet waste, an IR Sensor to detect waste entry, and an Ultrasonic Sensor to monitor bin fill levels. It alerts users with a buzzer and voice message if incorrect waste is disposed. When the bin becomes full, an automatic notification is sent for collection. This system promotes proper waste segregation, prevents overflow, and supports a cleaner and sustainable environment.

PROBLEM STATEMENT

Improper waste segregation at the source leads to environmental pollution and inefficient recycling processes. Manual monitoring of waste bins often results in overflow and delayed collection. Lack of awareness and real-time alerts causes mixing of wet and dry waste. There is no efficient system to detect incorrect disposal instantly. Therefore, a smart and automated waste monitoring system is required to ensure proper segregation and timely waste management.

INTRODUCTION

Waste management has become a serious issue due to the increasing amount of garbage generated in urban and rural areas. Improper segregation of wet and dry waste leads to environmental pollution, health hazards, and inefficient recycling. Traditional waste collection systems depend on manual monitoring, which often causes bin overflow and delayed disposal. The Smart Waste Segregation Monitoring System uses Arduino Uno along with moisture, IR, and ultrasonic sensors to automate waste detection and monitoring. This system ensures proper segregation, provides alerts for incorrect disposal, and helps maintain a cleaner and sustainable environment.

PROPOSED SYSTEM

- The system uses Arduino Uno as the main controller to manage all sensor operations and alerts.
- A Moisture Sensor is used to detect and differentiate wet waste from dry waste.
- An IR Sensor detects when waste is dropped into the bin and activates the checking process.
- An Ultrasonic Sensor monitors the fill level of the bin to prevent overflow.
- A buzzer and alert system notify users for incorrect disposal and inform authorities when the bin is full.
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PROCESS

01

When waste is dropped into the bin, the IR sensor detects the object and activates the system for analysis.

02

The moisture sensor checks the moisture content to determine whether the waste is wet or dry. If incorrect waste is detected, a buzzer or voice alert notifies the user.

03

The ultrasonic sensor continuously monitors the bin’s fill level, and when it reaches maximum capacity, an alert is sent for timely waste collection.

RESULT

The Smart Waste Segregation Monitoring System successfully detects wet and dry waste using moisture, IR, and ultrasonic sensors with accurate real-time monitoring. The system provides instant alerts for incorrect disposal and prevents overflow, ensuring efficient and smart waste management.

ANALYSIS

The system demonstrates reliable performance in accurately identifying waste type and monitoring bin levels using sensor-based automation. It reduces human intervention, improves segregation efficiency, and enhances overall waste management effectiveness.

CONCLUSION

The Smart Waste Segregation Monitoring System provides an efficient solution to improper waste management using Arduino Uno and sensor technology. It accurately identifies wet and dry waste through moisture detection and ensures proper disposal with instant alerts. The ultrasonic sensor prevents overflow by continuously monitoring the bin level. This system reduces manual effort and improves waste collection efficiency. Overall, the project contributes to cleaner surroundings and supports sustainable waste management practices.

THANK YOU